

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = (x + 7)^2 + 4$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

Correct Answer: -7

Rationale

The correct answer is -7 . For a quadratic function defined by an equation of the form $f(x) = a(x - h)^2 + k$, where a , h , and k are constants and $a > 0$, the function reaches its minimum when $x = h$. In the given function, $a = 1$, $h = -7$, and $k = 4$. Therefore, the value of x for which $f(x)$ reaches its minimum is -7 .